Department of Engineering Science
School of Science & Technology



CES 440 Data Communications
Ch.5 The Network Layer: Part I

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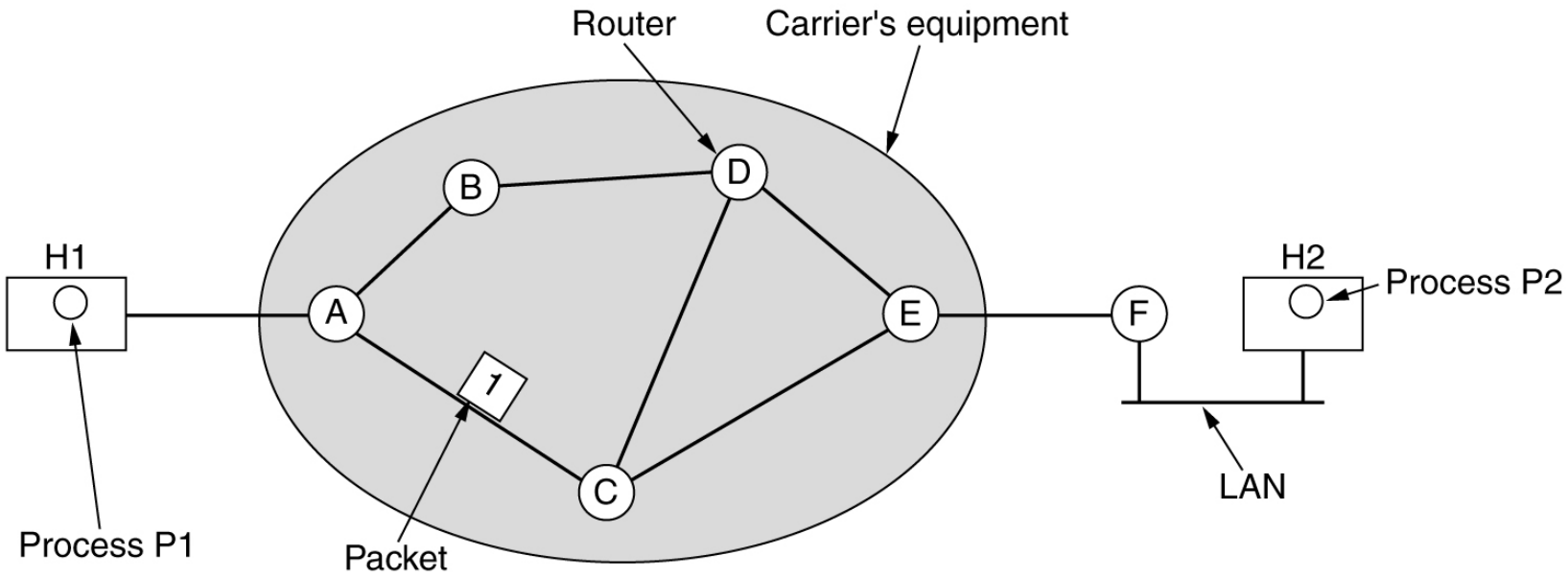
OUTLINE

- **Design Issues**
- **Routing Algorithms**
- **Internetworking**
- **Quality of Service**
- **Network Layer in the Internet**

DESIGN ISSUES: FUNCTIONS

- **Functions: getting packets from the source to the destination.**
  - Routing
    - choose appropriate path; knowledge of the topology.
  - Internetworking
    - connect different types of networks.
  - Quality of Service
  - Congestion Control
    - avoid overloading some of the lines.
  - Service to transport layer
    - Connectionless service
    - Connection oriented service (virtual circuit service)
    - Transport layer should be shielded from the number, type, and topology of the routers present.

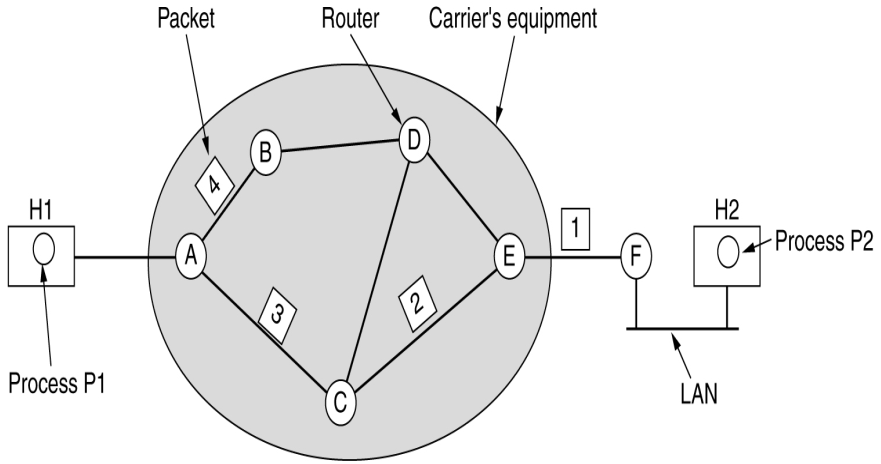
DESING ISSUES: OPERATION CONTEXT



Routers can be in either subnet (owned by operator) or LAN (owned by end users).

DESING ISSUES: CONNECTIONLESS SERVICES

- **Connectionless Service**
  - Datagram: packet in connectionless network.
  - Datagram subnet: connectionless subnet.
  - Each datagram is routed individually.
    - Can take different route.
  - Store-and-forward:
    - Packet received by router in its entirety.
    - Packet stored in router, checksum examined by router.
    - Packet is forwarded to the next router.



A's table

Dest.	initially	later
A	–	–
B	B	B
C	C	C
D	B	B
E	C	B
F	C	B

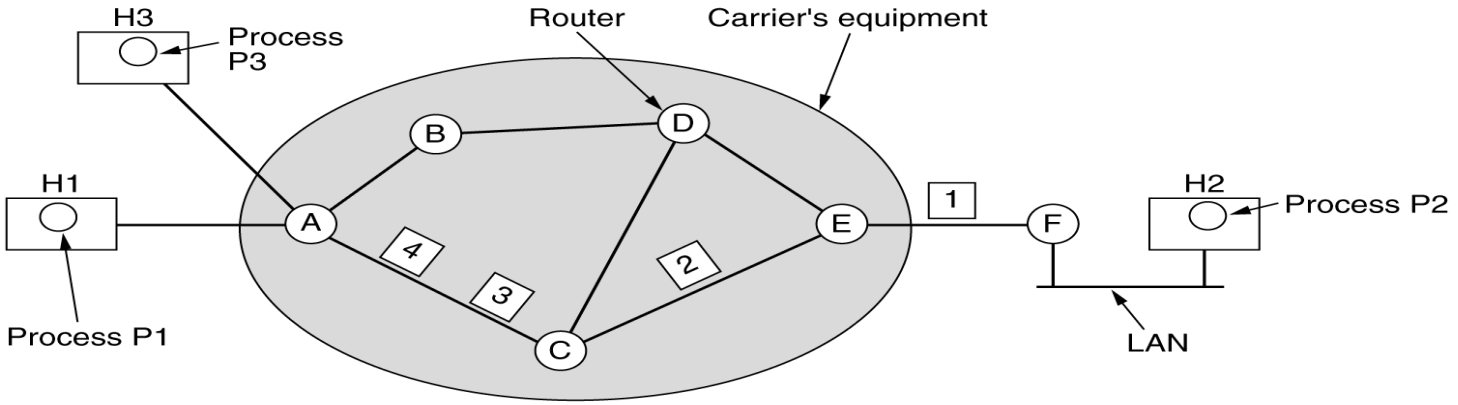
Dest.	C's table
A	A
B	A
C	–
D	D
E	E
F	E

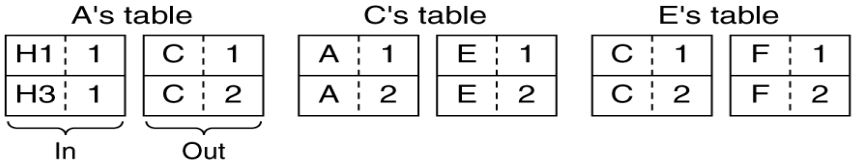
Dest.	E's table
A	C
B	D
C	C
D	D
E	–
F	F

Dest. Line

DESIGN ISSUE: VIRTUAL CIRCUIT SERVICE

- **Connection in network layer is called virtual circuit (VC)**
  - When a connection is setup, a virtual circuit (route from source to destination) is chosen.
    - All packets will follow the same route (This is only true for network layer!).
    - The subnet is called virtual circuit subnet.
    - Each packet carries an VC identifier.
- **Connection Oriented ≠ Virtual circuit**
  - In other layers, e.g., application layer, transport layer, packets belong to the same connection might tack different route!!!





DESIGN ISSUE: DATAGRAM V.S. VC

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

OUTLINE

- Design Issues
- **Routing Algorithms**
- Internetworking
- Quality of Service
- Network Layer in the Internet

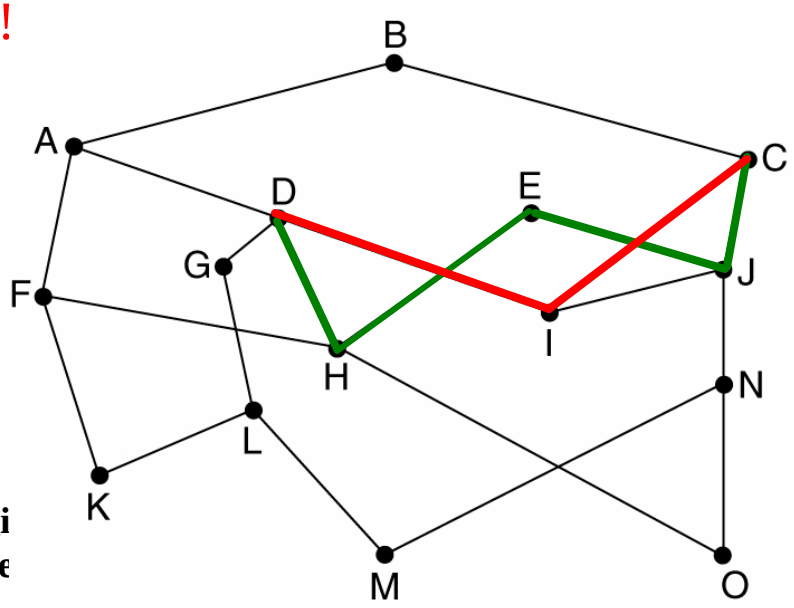
ROUTING: GENERAL CONCEPTS

- **Routing Algorithm**
  - Deciding which output line an incoming packet should be forwarded to.
  - Datagram: routing is performed for every packet.
  - VC: routing is only performed at connection setup ➔ session routing.
- **Routing v.s. Forwarding**
  - Forwarding: send the packet to the next router.
  - Routing: decide the best route by filling in and updating the routing table.
- **Static Routing v.s. Adaptive Routing**
  - Static routing: the choice of the route is fixed
  - Dynamic routing: dynamically choose route based on topology, traffic, etc.
- **Measures**
  - Throughput, Delay, Fairness,
  - # of hops, distance

ROUTING: OPTIMALITY PRINCIPLE

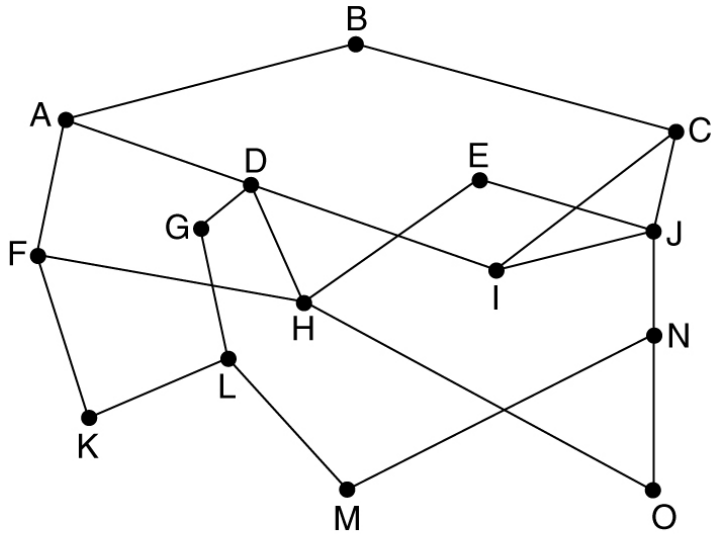
- **Optimality Principle**
  - If router D is on the optimal path from A to C, then the optimal path from D to C must falls along the same route.
  - Proof by contradiction:

    1. Optimum route from A to C: A-D-I-C.
    2. Assume D-H-E-J-C is the optimum route between D and C.
    3. Then D-H-E-J-C will have a lower cost than D-I-C.
    4. A-D-H-E-J-C will have a lower cost than A-D-I-C. ➔ Contradiction!

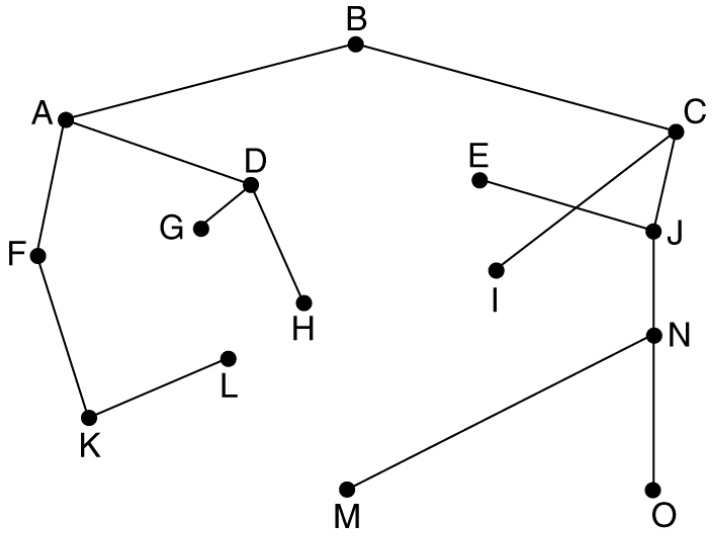


ROUTING: OPTIMALITY PRINCIPLE

- **Sink Tree**
  - Based on the optimality principle, the set of optimal routes from all sources to a given destination form a tree rooted at the destination ➔ sink tree.
  - These is no loops in the sink tree.
  - Routing algorithm: find out the sink tree!



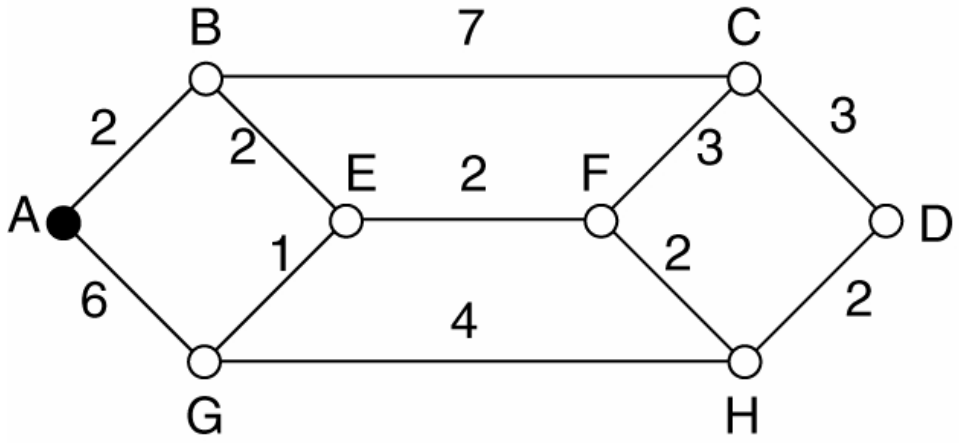
(a)



(b)

ROUTING: SHORTEST PATH ROUTING

- **Shortest Path: find out the route with the shortest "distance" between source and destination.**
  - Different measures can be used to measure "distance"
    - E.g. # of hops, physical distance (km), queuing delay …
  - In practical system, the "distance" measure is usually a function of the distance, bandwidth, average traffic, communication cost, mean queue length, delay, and other factors.

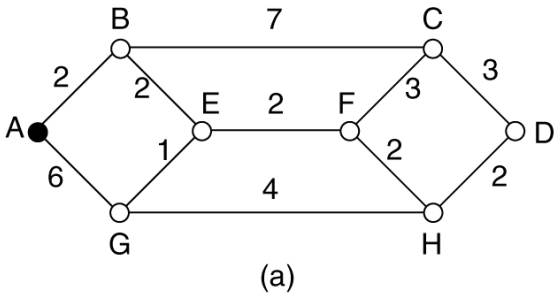


(a)

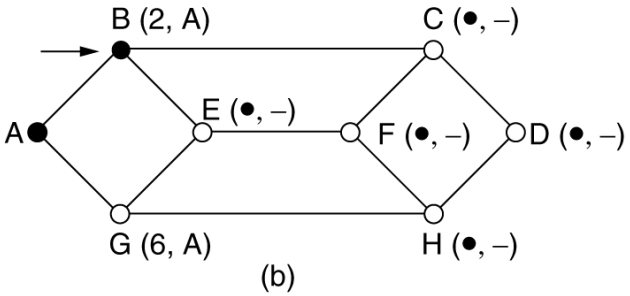
ROUTING: SHORTEST PATH ROUTING

Find the shortest path from A to D (Dijkstra Algorithm)

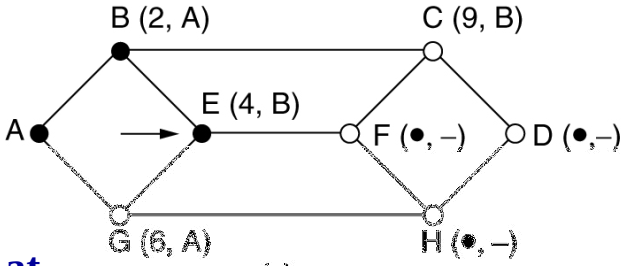
- ● **Permanent node**
- ○ **Tentative node**



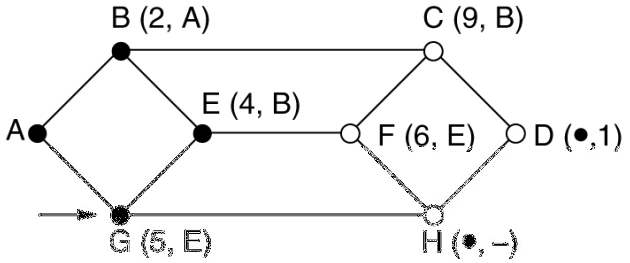
(a)



(b)

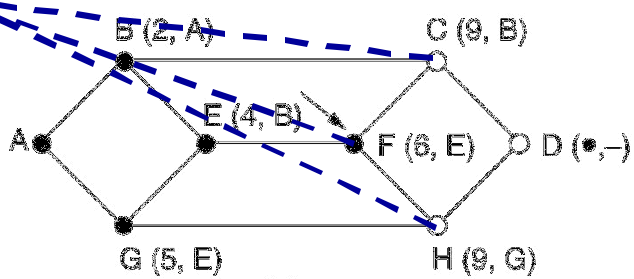


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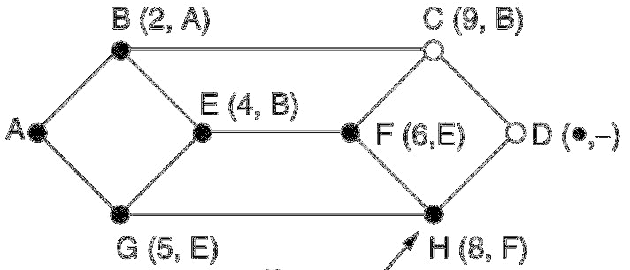


(d)

Comparing the cost at all tentatively labeled nodes.



(e)



(f)

ROUTING: FLOODING

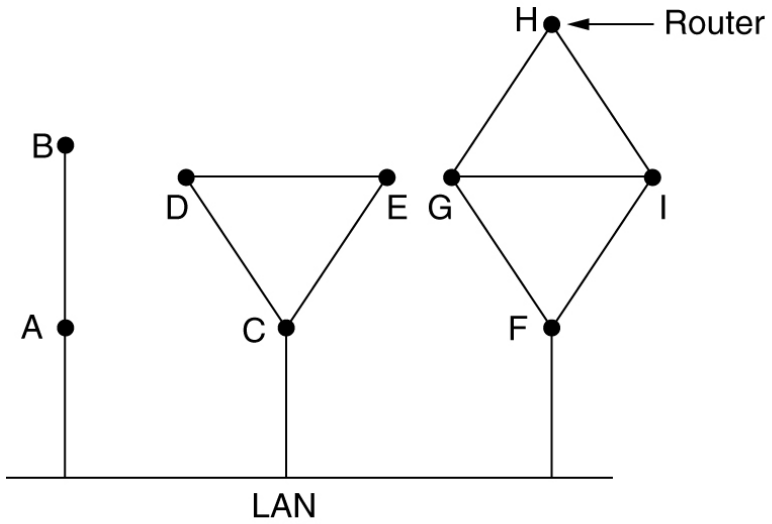
- **Flooding (static routing algorithm)**
  - Every incoming packet is sent out on every outgoing line except the one it arrived on.
  - Problem: Generate infinite number of packets.
  - Solution 1: Each packet header contains a hop counter.
    - Counter decreases by 1 after each hop.
    - Packet is discarded after its counter reaches zero.
  - Solution 2: Each router keeps track of each packet, and only flood new packets.
    - Each source router puts a sequence number in the packet header.
    - Each router keeps one list per source router telling which packets from that source router has been flooded already.
      - For each source router, all the other router needs to keep one list for it.
- **Selective Flooding**
  - Router only sends out packet on the lines that are going approximately in the right direction based on its knowledge of the network topology.
- **Applications**
  - Systems requiring high reliability. E.g. military applications
  - Can be used as a metric to evaluate other algorithms.

ROUTING: LINK STATE ROUTING

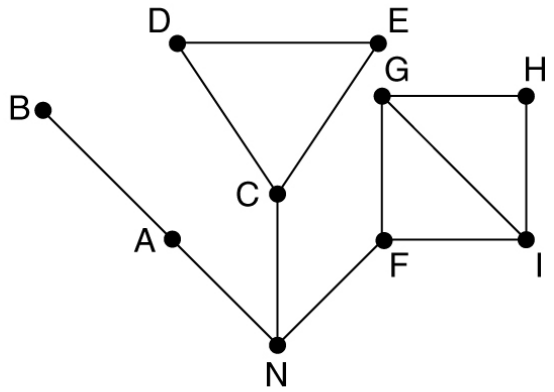
- **Link State Routing**
  - Used in Internet
  - Five Steps
    - Learning about neighbors
    - Measuring line costs to neighbors
    - Building link state packets
    - Distributing link state packets
    - Computing the network routes.

ROUTING: LINK STATE ROUTING

- **Learning about neighbors**
  - When router boots up, send a HELLO packet to each P2P line.
  - The router on the other end sends a REPLY telling who it is.
  - LAN is usually represented as an independent node.



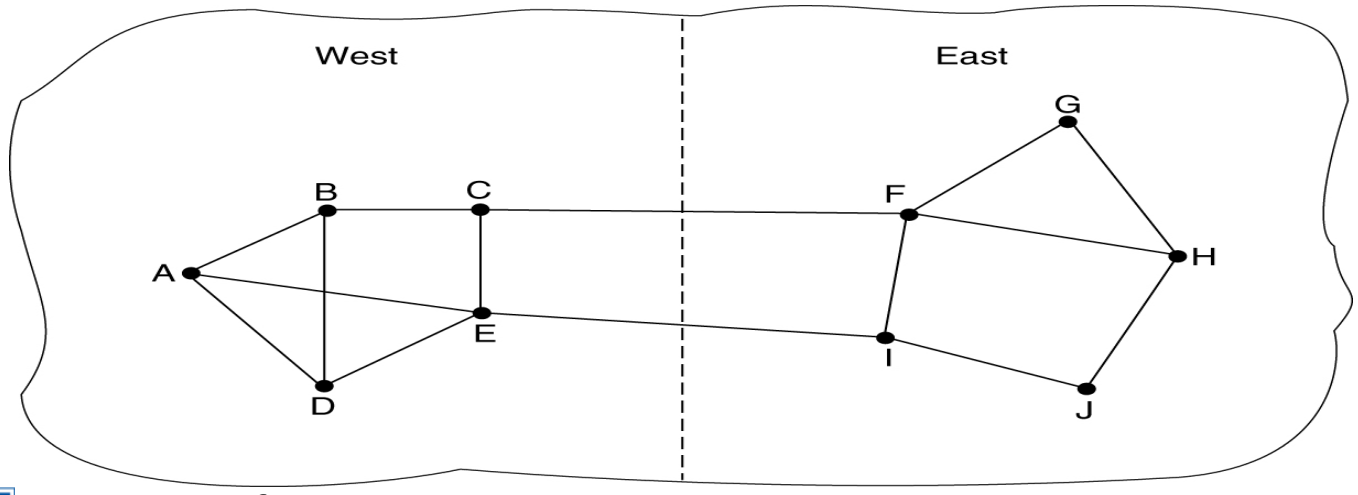
(a)



(b)

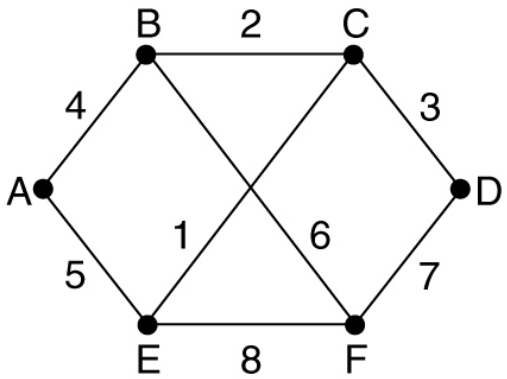
ROUTING: LINK STATE ROUTING

- **Measuring line cost**
  - Send an ECHO packet to neighbor
  - The neighbor is required to reply immediately.
  - Measuring the round trip delay of the line.
  - To take into consideration of traffic load of the source router
    - The timer is started when the ECHO packet is queued in source.
  - To ignore the traffic load of the source router
    - The timer is started when the ECHO packet reaches front of queue.



ROUTING: LINK STATE ROUTING

- **Building link state packets**
  - Each router will send out a link state packet periodically.
  - Link state packet contains {Source ID, Seq#, Age, (neighbor ID, distance to neighbor)}.



(a)

Link State Packets

A	B	C	D	E	F
Seq.	Seq.	Seq.	Seq.	Seq.	Seq.
Age	Age	Age	Age	Age	Age
B 4	A 4	B 2	C 3	A 5	B 6
E 5	C 2	D 3	F 7	C 1	D 7
	F 6	E 1		F 8	E 8

(b)

If one router collects all the link state packets from all the other routers, the router will have full knowledge of network topology!

ROUTING: LINK STATE ROUTING

- **Distributing the link state packets**
  - Problem:
    - when a router gets a new link state packet, it will update the network topology in memory.
    - Different routers might have different versions of topology.
      - Inconsistencies, loops, unreachable machines, etc.
    - How to reliably deliver link state packets?
  - Solution: Flooding
    - Each packet is assigned a sequence number by source router.
    - Each router maintains a list per source router to record which packet has been flooded.
      - Only record the seq.# of the last packet flooded.
      - If the seq.# of an arriving packet is smaller than the seq.# in list, the packet was flooded before and will be discarded.
        - » (A, 4) is in list. If (A, 3) arrives, it will be discarded.

ROUTING: LINK STATE ROUTING

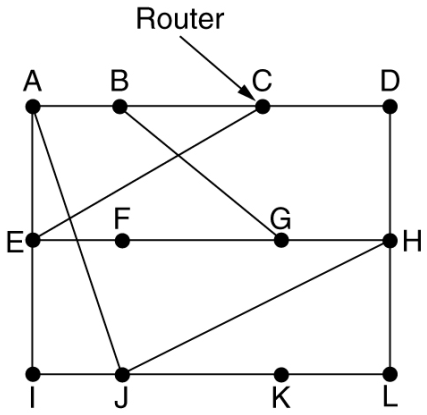
- **Distributing the link state packets (Cont'd)**
  - Problems for flooding with seq.#
    - Seq.# will wrap around. (e.g. 2-bit seq#: 0 1 2 3 0 1 2 3 0 1 2 3 …)
    - If a router crashes, it will send link state packet starting from 0.
    - Seq.# could crash
  - Solutions: including an age field.
    - If the age hits zero, the information from that link state packet will be discarded ➔ the entries in the status list will be cleared ➔ the router is ready to accept any new packet from the particular source router.
  - To improve reliability
    - Link state packets will be put in a holding area before Tx
      - If a new link state packet comes in, the one in holding will be discarded.
    - Each link state packet is acknowledged by the receiver.

ROUTING: LINK STATE ROUTING

- **Computing the new routes**
  - Once a router has collected link state packets from all routers in the network, it has full knowledge of the subnet topology.
  - Dijkstra's algorithm used in Shortest Path Routing can be used to find the shortest path between any two routers.
  - If the subnet contains *n* routers, each with *k* neighbors
    - Each router needs *nk* memory blocks to store all the link state packets.
- **Example Link State Routing protocols**
  - IS-IS (Intermediate System-Intermediate System)
    - Adopted by ISO for its connectionless network layer
    - Used in some Internet backbones (old NSFNET)
  - OSPF (Open Shortest Path First)
    - Widely used in Internet.

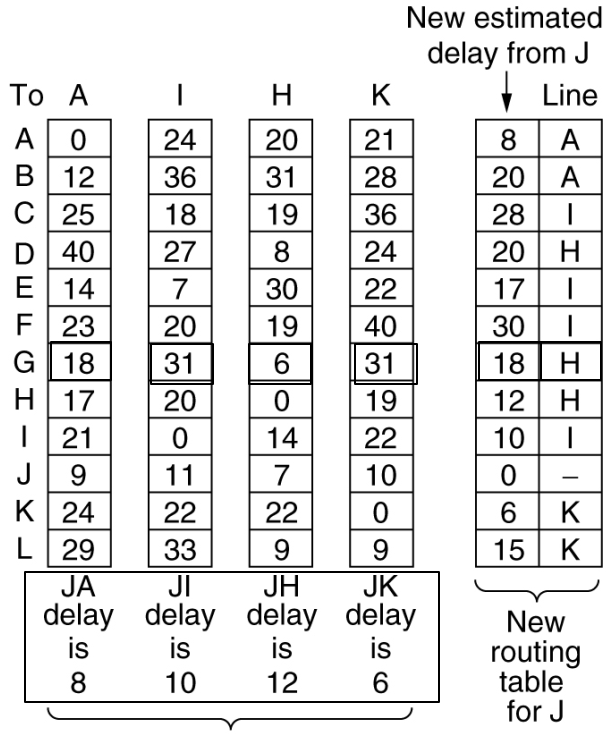
ROUTING: DISTANCE VECTOR ROUTING

- **Distance Vector Routing**
  - Each router maintain a routing table (a vector) contains one entry for each destination router as: (destination, distance to destination, which line to use).
  - The contents of the tables are dynamically updated by exchanging information with neighboring routers.
  - Example: How router J computes its routing table.



(a)

J→A: 8, A→G: 18. J→A→G: 8+18=26.
J→I: 10, I→G: 32. J→I→G: 10+32=42.
J→H: 12, H→G: 6. J→H→G: 12+6=18.
J→K: 6, K→G: 31. J→K→G: 6+31=37.

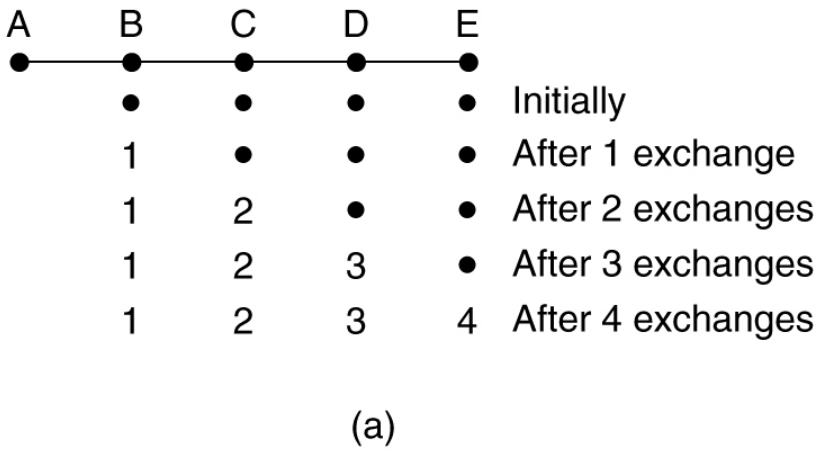


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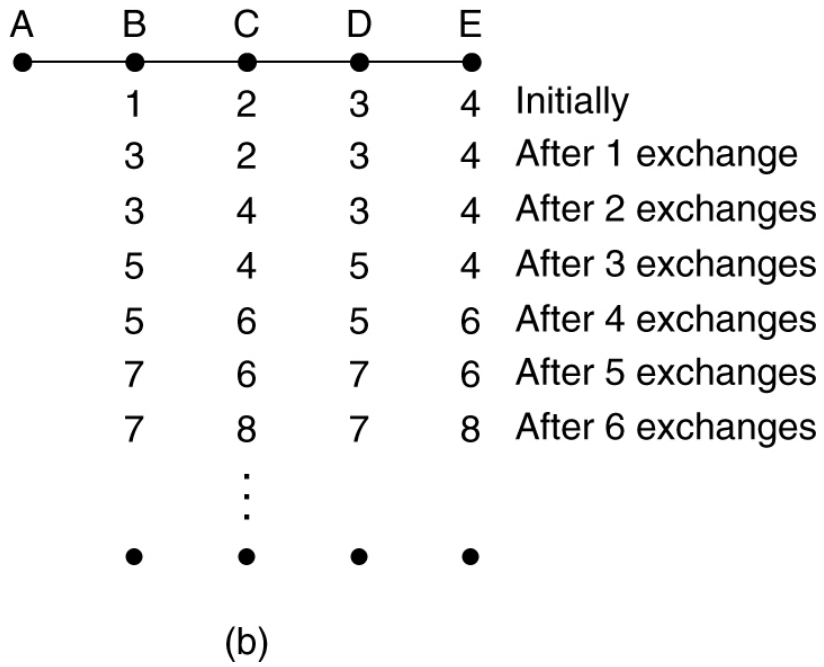
ROUTING: DISTANCE VECTOR ROUTING

- **Count-to-infinity problem**
  - Example: the entries of destination A in the routing tables of router B, C, D, and E. (distance = hop)

When A boots up



After A breaks down

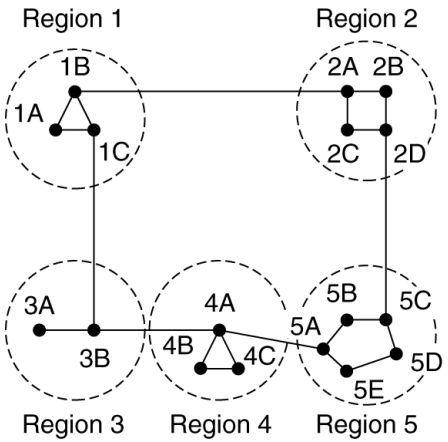


ROUTING: HIERARCHICAL ROUTING

- **Motivation:**
  - Routing table size grows proportionally with network size.
    - More memory, more CPU time, and more bandwidth are required for routing.
- **Solution: Hierarchical Routing**
  - Coverage area are divided into regions.
  - Each router only knows the details of the region it belongs to.
    - It doesn't know the detailed structure of other regions.
  - Larger regions can be further divided into smaller regions
    - Multi-level hierarchy.

ROUTING: HIERARCHICAL ROUTING

- **Example:**
  - With two-layer hierarchy, # of entries of routing table of 1A reduced from 17 to 7.
  - Price: increase in path length
    - 1A – 5C: shortest path is from region 2.
    - Region 3 is used instead for hierarchical routing.



(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

ROUTING: HIERARCHICAL ROUTING

- **How to calculate the # of entries in routing table?**
  - Consider a subnet with 720 routers
  - No hierarchy:
    - each router needs to have 720 entries
  - 2-layer hierarchy with 24 regions, 30 routers/region
    - Each router needs 30 local entries plus 23 remote entries:

      30 + 23 = 53 entries.
  - 3-layer hierarchy with 8 clusters, 9 regions/cluster, 10 routers/region
    - Each router needs 10 local entries, 8 entries for other regions within the same cluster, 7 entries for distant clusters

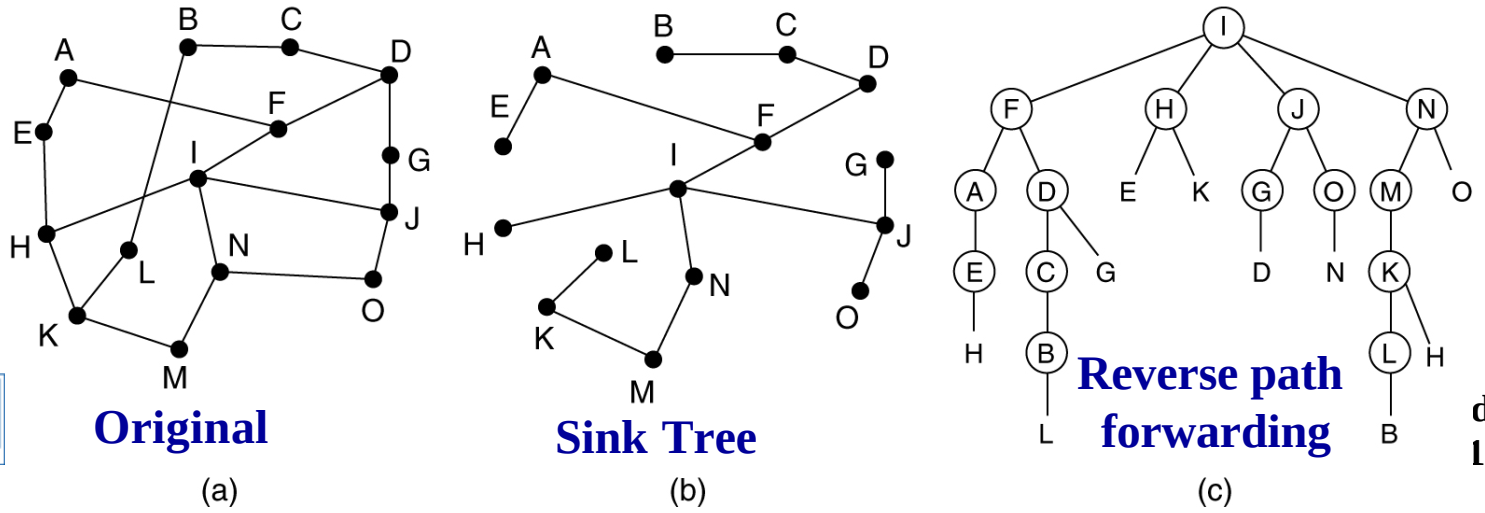
      10 + 8 + 7 = 25 entries
- **It was discovered that optimal levels for an *N*-router subnet is ln*N*.**

ROUTING: BROADCAST ROUTING

- **Broadcast routing:**
  - sending a packet to all destinations simultaneously. (e.g. live radio)
- **Method 1:**
  - source sends packet to each destination individually.
- **Method 2: flooding**
- **Method 3: multi-destination routing**
  - The broadcast packet carries a list of destinations.
  - Each router checks the destination lists and determines the corresponding outgoing line.
    - A new copy of the packet with only corresponding destinations is generated for each outgoing line.
    - After sufficient number of hops, each packet will carry only one destination.

ROUTING: BROADCAST ROUTING

- **Method 4: spanning tree**
  - Spanning tree: a graph contains all routers but no loops.
  - A spanning tree can be built with the source router being root.
  - Generates the absolute minimum number of packets (most efficient)
  - Each router needs to have some knowledge of spanning tree.
- **Method 5: reverse path forwarding**
  - When a broadcast packet arrives, the router checks its routing table to see if it arrives from the line used to send packet *to* the source.
  - If it is, the packet is forwarded to all the other lines except the incoming line. Otherwise packet is discarded.
    - E.g. routing entry in A (I, Line FA). If packet with source I arrives from Line FA, it will be forwarded.



ROUTING: BROADCAST ROUTING

- **Method 5: reverse path forwarding (Cont'd)**
  - Example: entries for dest. I in the routing tables of all routers

Dest	Next
I	F

Router A

Dest	Next
I	C

Router B

Dest	Next
I	D

Router C

Dest	Next
I	F

Router D

Dest	Next
I	A

Router E

Dest	Next
I	I

Router F

Dest	Next
I	J

Router G

Dest	Next
I	I

Router H

Dest	Next
I	I

Router I

Dest	Next
I	I

Router J

Dest	Next
I	M

Router K

Dest	Next
I	K

Router L

Dest	Next
I	N

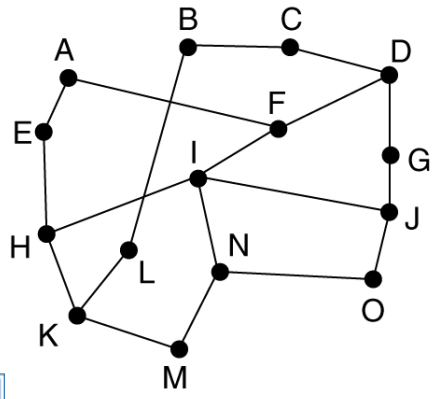
Router M

Dest	Next
I	I

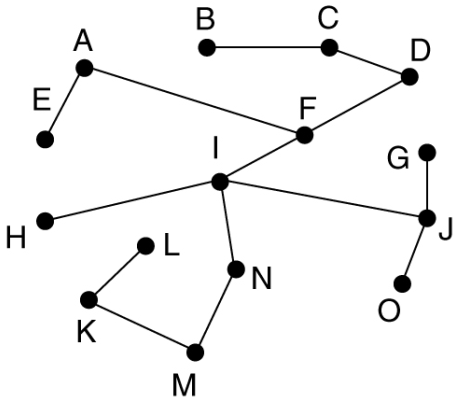
Router N

Dest	Next
I	J

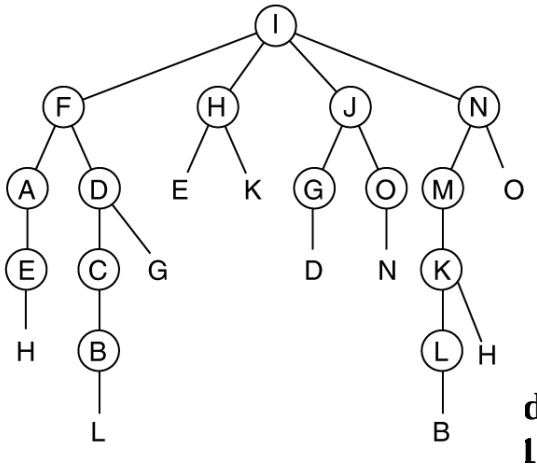
Router O



(a)



(b)



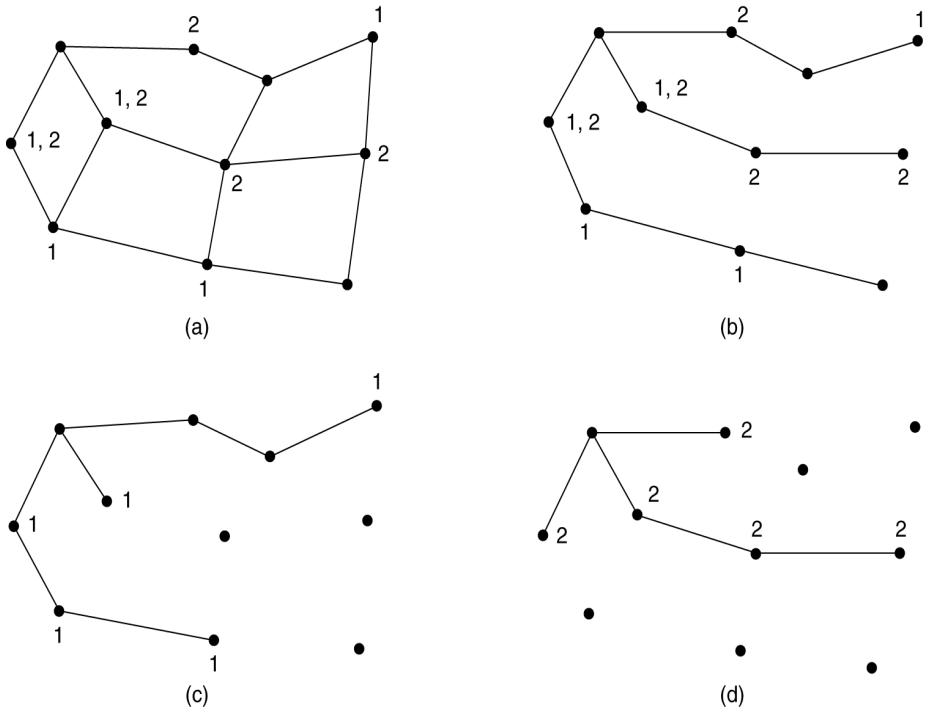
(c)

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16/2005

ROUTING: MULTICAST ROUTING

- **Multicasting**
  - Sending a packet to a group of hosts.
  - Requires group management: each router needs to know which hosts belonging to which group.
  - Each router computes a spanning tree covering all other routers.
  - The spanning tree is pruned by removing all irrelevant lines.
- **Alternative: Core-based trees**
  - A single spanning tree for each group is computed, and it's stored in a router close to center of the group (core).
  - Multicast message is first sent to core.



ROUTING: ROUTING FOR MOBILE HOSTS

- **Routing for mobile hosts**
  - World is divided into small areas (cells)
    - Each mobile host designate one area as home.
    - Home agent: keeps track of all hosts with home in this area
    - Foreign agent: keeps track of hosts currently in this area, but with home in other areas
  - When a host enters an area, it must register with foreign agent in that area.
    - Foreign agent will notify the mobile host's home agent about its current location.
  - Packets to mobile hosts will first be sent to its home agent, then forwarded to its current location.
    - Home agent will notify sender to send following packets to host's current location.
  - Example: cell phone system.

ROUTING: ROUTING FOR MOBILE HOSTS

